

Loops 1.1

■ Basic

1. Print numbers from 1 to 10 using a for loop.
2. Print numbers from 10 down to 1 using a while loop.
3. Print the multiplication table of any number entered by the user.
4. Print all even numbers between 1 and 50.
5. Calculate the sum of numbers from 1 to 100 using a loop.
6. Print the squares of numbers from 1 to 10.
7. Print each character of a string using a loop.
8. NEW: Print numbers from 1 to 10, but stop the loop if the number reaches 7 (break).
9. NEW: Print numbers from 1 to 10, but skip printing 5 (continue).
10. NEW: Print numbers 1–5, and after the loop finishes, print 'Loop completed' using else.

■ Intermediate

1. Print the factorial of a given number using a loop.
2. Reverse a string using a loop.
3. Count the number of vowels in a given string using a loop.
4. Print the Fibonacci series up to n terms using a loop.
5. Print all prime numbers between 1 and 50 using loops and conditionals.
6. Find the largest element in a list using a loop (without max()).
7. Print only positive numbers from a list using a loop.
8. NEW: Keep asking the user for numbers until they enter a negative number (while + break).
9. NEW: Search for a character in a string using a loop. If found, break and print 'Found', else print 'Not Found' using else.
10. Write a program to count frequency of each character in a string using loops.

■ Advanced

1. Print a pyramid pattern of stars (*) using nested loops.
2. Print an inverted pyramid pattern of stars using nested loops.
3. Print a multiplication table from 1 to 10 using nested loops.
4. Print a right-angled triangle of numbers using nested loops.
5. Print the sum of digits of a number using a loop.
6. Check if a number is an Armstrong number using a loop.
7. NEW: Search for a number in a list. Stop when found (break), else print 'Not Found'.
8. NEW: Print prime numbers between 1–50, but skip multiples of 5 (continue).
9. NEW: Simulate login attempts: allow 3 tries, exit loop if successful (break), else print 'Account Locked' using else.

10. NEW: Check if a list is sorted. If unsorted, break immediately; else print 'List is sorted' using else.